Partitioning Israeli Municipalities into Politically Homogeneous Cantons: A Constrained Spatial Clustering Approach

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M.Sc. in Computer Science (Project Track), Bar-Ilan University, 2025-2026

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Abstract

Israeli society has experienced significant political polarization in recent years, reflected in five Knesset elections held within a four-year period (2019-2022). Public discourse increasingly references hypothetical divisions of the country into politically homogeneous "cantons." This project develops a data-driven algorithmic approach to explore such divisions using publicly available municipality-level election results and geographic boundary data from the Israel Central Bureau of Statistics (CBS).

We partition 229 Israeli municipalities into geographically contiguous cantons that maximize internal political similarity. Our methodology employs four clustering algorithms - Simulated Annealing, Agglomerative Clustering with contiguity constraints, Louvain Community Detection, and K-Means (baseline) - evaluated across four feature representations (BlocShares, RawParty, PCA, NMF), three distance metrics (Euclidean, Cosine, Jensen-Shannon), and six values of K (3-20), yielding a comprehensive grid of experimental configurations.

Key results show that the BlocShares representation with Euclidean distance produces the highest clustering quality (silhouette score 0.905), while Simulated Annealing achieves the best population balance among contiguity-respecting algorithms. Temporal stability analysis across all five elections reveals that deterministic algorithms (Louvain, Agglomerative) produce near-perfectly stable partitions (ARI up to 1.0), while Israel's political geography remains structurally consistent despite electoral volatility. The resulting K=5 partition identifies five politically coherent regions: a right-wing southern periphery, a right-dominant urban center with significant Haredi presence, a center-leaning coastal belt, a left-leaning Druze enclave, and a large Arab-majority northern bloc - closely reflecting known political-demographic divisions. Comparison with Israel's administrative districts yields an Adjusted Rand Index of 0.435, confirming that political cantons follow ideological rather than ad-

ministrative boundaries. These findings provide a data-driven framework for understanding Israel's latent political geography and demonstrate the applicability of constrained spatial clustering to politically polarized societies.

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1. Introduction

1.1 Motivation

Over the past four years, Israel held five Knesset elections (April 2019, September 2019, March 2020, March 2021, and November 2022), an unprecedented frequency that reflects deep political divisions across multiple dimensions: secular versus religious, left versus right on the ideological spectrum, Arab versus Jewish communal politics, and attitudes toward judicial reform and governance. Throughout this period, public and media discourse repeatedly invoked the notion of dividing Israel into separate "cantons" along political lines - contrasting, for example, Tel Aviv's liberal-secular majority with Jerusalem's religious-conservative population, or the Arab-majority towns of northern Israel with predominantly Jewish communities elsewhere.

While such discourse is largely rhetorical, it raises a compelling computational question: if Israel were to be divided into politically coherent regions based purely on voting patterns, what would those regions look like, and how stable would they be across election cycles? This project applies data-driven computational methods to answer this question rigorously.

1.2 Research Objectives

This project has three primary objectives:

1. **Partition 229 Israeli municipalities into K geographically contiguous cantons** that maximize internal political homogeneity, using municipality-level election results from five Knesset elections.
2. **Systematically compare clustering approaches** across multiple feature representations, distance metrics, algorithms, and values of K, evaluating trade-offs between political homogeneity, population balance, and geographic compactness.
3. **Analyze the temporal stability** of canton boundaries across election cycles to determine whether Israel's political geography is structurally persistent or volatile.

1.3 Contributions

This project makes the following contributions:

- A novel application of constrained spatial clustering to Israeli political geography, combining graph-based contiguity constraints with political feature-space optimization.
- A comprehensive experimental framework evaluating 264 configurations across four representations, three distance metrics, four algorithms, and six values of K.
- Temporal stability analysis across five elections using Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI).

- Qualitative case studies validating that algorithmic cantons align with known political-demographic divisions.
- An interactive Streamlit web application for exploring canton partitions, experiment results, and stability analysis.
- An open-source, modular Python codebase enabling reproducible analysis and extension.

1.4 Report Outline

Section 2 reviews related work on electoral redistricting, spatial clustering, and Israeli political geography. Section 3 formally defines the constrained canton partitioning problem. Section 4 describes the data sources and processing pipeline. Section 5 presents the methodology - feature representations, distance metrics, graph construction, clustering algorithms, and evaluation metrics. Section 6 reports experimental results from the 264-configuration grid search. Section 7 analyzes temporal stability across elections. Section 8 presents qualitative case studies. Section 9 discusses limitations and future work, and Section 10 concludes.

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2. Literature Review

2.1 Electoral Redistricting and Gerrymandering

The problem of dividing a territory into politically meaningful regions has deep connections to electoral redistricting research. In democratic systems with district-based representation, electoral boundaries must be periodically redrawn to reflect population changes, and the drawing process can be manipulated to favor particular parties - a practice known as gerrymandering [1].

Brieden et al. (2017) proposed a constrained clustering approach to electoral redistricting based on generalized Voronoi diagrams and linear programming [2]. Their method partitions geographic units into districts with hard population balance constraints while minimizing intra-district political heterogeneity. The key innovation is formulating the problem as a continuous optimization over diagram parameters, enabling efficient computation even for large instances.

Our project differs from classical redistricting in important ways: (1) we aim to *maximize* political homogeneity within cantons, whereas fair redistricting often seeks competitive or balanced districts; (2) we operate at the municipality level rather than individual census blocks; and (3) our work is purely descriptive and exploratory rather than prescriptive. However, the shared technical challenges - ensuring geographic contiguity, handling population constraints, and operating on spatial graphs - make redistricting research directly relevant.

Recent computational approaches to fair redistricting include Markov Chain Monte Carlo (MCMC) methods that sample from the space of valid partitions to detect gerrymandering outliers [3], and graph-based metrics for measuring district compactness [4]. These methods underscore the difficulty of the constrained partitioning problem and motivate the multi-algorithm comparison in our experimental framework.

2.2 Spatial and Constrained Clustering

Standard clustering algorithms - K-Means, hierarchical agglomerative methods, spectral clustering - operate in feature space without regard to geographic structure [5]. When geographic contiguity is required (as in regionalization), these methods must be extended with spatial constraints.

Spatially-constrained agglomerative clustering is a natural approach: starting with each spatial unit as its own cluster, iteratively merge adjacent clusters that are most similar in feature space. This maintains contiguity by construction, as merging two connected clusters yields a connected cluster [6]. Ward's criterion, which minimizes the increase in within-cluster sum of squares at each merge, is commonly used for the merge cost function.

Spectral clustering leverages the graph Laplacian to embed graph nodes into a low-dimensional space before applying K-Means [7]. When applied to a contiguity graph with edge weights derived from political similarity, spectral methods can discover clusters that respect both graph topology and feature-space structure.

Community detection algorithms from network science, particularly the Louvain method [8], offer an alternative by maximizing modularity - a measure of the fraction of edges within communities versus between communities. While Louvain does not directly control the number of clusters K, it provides a topology-driven baseline partition. Louvain operates on a similar principle to spectral clustering (leveraging graph structure rather than feature-space distances) and was selected over spectral methods for our experiments due to its computational efficiency and natural handling of weighted adjacency graphs.

Metaheuristic approaches such as Simulated Annealing (SA) can optimize arbitrary multi-objective cost functions over the partition space while maintaining contiguity through constrained neighborhood operations [9]. SA is well-suited to this problem because it can balance competing objectives (homogeneity, population balance, compactness) through a weighted cost function and escape local optima through stochastic acceptance of worse solutions.

2.3 Political Geography of Israel

Israel's political landscape is characterized by several cross-cutting cleavages [24]:

- **Left-Right ideological spectrum:** The security-oriented right (Likud, Religious Zionism) versus the peace-process-oriented left (Labor, Meretz).
- **Secular-Religious divide:** Secular and traditional voters versus ultra-Orthodox (Haredi) parties (Shas, United Torah Judaism).
- **Arab-Jewish divide:** Arab citizens voting predominantly for the Joint List or Ra'am versus Jewish-majority parties.
- **Geographic concentration:** These cleavages exhibit strong spatial autocorrelation - nearby municipalities tend to vote similarly due to shared demographics, economic conditions, and social networks [10, 25, 30].

The geographic structure of Israeli politics makes it a suitable testbed for constrained spatial clustering: we expect meaningful, contiguous cantons to emerge that reflect these known political-demographic divisions.

2.4 Dimensionality Reduction for Voting Data

Municipality-level election data can be represented as high-dimensional vectors of party vote shares. Dimensionality reduction techniques compress these vectors while preserving key political structure:

- **Principal Component Analysis (PCA)** finds orthogonal axes of maximum variance, typically capturing left-right and Arab-Jewish dimensions in the first few components [11].
- **Non-negative Matrix Factorization (NMF)** decomposes vote shares into additive components, producing interpretable "political archetypes" (e.g., an Arab voting pattern, a Haredi pattern) [12].
- **Manual bloc aggregation** groups parties into predefined political blocs (Right, Haredi, Center, Left, Arab), producing a low-dimensional but domain-informed representation.

We systematically compare these representations to assess which best captures politically meaningful variation for clustering purposes.

2.5 Research Gap

Prior work on constrained spatial clustering has focused primarily on fair electoral redistricting – producing *competitive* districts with balanced partisan composition [1, 2, 3]. In contrast, our work seeks to maximize political *homogeneity* within regions, a fundamentally different objective. While regionalization methods such as SKATER [23] and MAX-P-REGIONS [29] address spatially constrained partitioning, they have not been applied to Israeli political geography, nor have they been evaluated across a systematic grid of feature representations, distance metrics, and algorithms. Furthermore, existing studies of Israeli political geography [10, 25] are primarily qualitative, lacking the quantitative spatial clustering framework that our project provides. This work bridges these gaps by combining constrained spatial clustering with a comprehensive experimental comparison tailored to the Israeli political context.

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3. Problem Definition

3.1 Formal Statement

Let $V = \{v_1, \dots, v_n\}$ be a set of n municipalities, each characterized by a political feature vector $f_i \in \mathbb{R}^d$ derived from election results. Let $G = (V, E)$ be a geographic contiguity graph where edge (v_i, v_j) exists if municipalities i and j share a geographic boundary. Let w_i denote the voter population of municipality i.

The Canton Partitioning Problem: This is a variant of the regionalization problem [23, 29] – given V, G, feature vectors $\{f_i\}$, voter weights $\{w_i\}$, and a target number of cantons K, find a partition $\mathcal{C} = \{C_1, \dots, C_K\}$ of V that:

1. **Maximizes political homogeneity:** Minimizes within-canton political variance.

2. **Ensures geographic contiguity:** Each canton C_k induces a connected subgraph of G.
3. **Balances population:** Minimizes disparity in total voter population across cantons.

3.2 Multi-Objective Formulation

These objectives may conflict: maximizing homogeneity may produce cantons with extreme population imbalance, while enforcing strict balance may force politically dissimilar municipalities together. We adopt a weighted multi-objective approach:

$$\text{Cost}(\mathcal{C}) = \alpha \cdot \text{Homogeneity}(\mathcal{C}) + \beta \cdot \text{Balance}(\mathcal{C}) + \gamma \cdot \text{Compactness}(\mathcal{C})$$

where α, β, γ are tunable weights. In our Simulated Annealing implementation, we use $\alpha = 0.4$, $\beta = 0.4$, $\gamma = 0.2$. We assign equal weight to homogeneity and balance as both are primary objectives, with compactness as a secondary regularizer. These weights were selected based on preliminary experiments; sensitivity to weight variation is noted in Section 9.

3.3 Contiguity Constraint

The contiguity constraint is fundamental to producing geographically meaningful cantons. It requires that each canton C_k forms a connected subgraph of G - i.e., any two municipalities in the same canton can be reached from each other by traversing edges within that canton. This constraint is enforced structurally in agglomerative methods (only adjacent clusters merge) and algorithmically in SA (moves are rejected if they break contiguity).

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4. Data Sources

4.1 Election Data

We use municipality-level election results from five consecutive Knesset elections, obtained from the Israel Central Bureau of Statistics (CBS) [26]:

Table 1: Municipality-level election data across five Knesset elections (all cover 229 municipalities).

Election	Knesset	Date	Eligible Voters	Actual Voters	Turnout
1	21	Apr. 2019	6,014,124	3,873,326	62.7%
2	22	Sep. 2019	6,061,316	3,950,538	65.5%
3	23	Mar. 2020	6,118,607	4,048,714	67.4%
4	24	Mar. 2021	6,232,307	3,781,640	59.1%
5	25	Nov. 2022	6,426,211	4,084,119	62.8%

All five elections cover the same 229 municipalities after name normalization and cross-election matching. Each record contains per-party vote counts, enabling computation of vote share vectors.

4.2 Geographic Data

Municipal boundary polygons were obtained from the Israel CBS geographic databases [27] in Shapefile format. Municipalities were dissolved (merged) from locality-level polygons to municipality-level using an official locality-to-municipality mapping. The resulting GeoJSON file contains 234 polygon features with Hebrew municipality names. Of these, 229 are consistently matchable across all five elections and form the analysis set; the remaining 5 appear in the geographic data but could not be matched across all election datasets.

4.3 Party-to-Bloc Mapping

Israeli political parties were classified into five political blocs:

Table 2: Party-to-bloc classification.

Bloc	Description	Example Parties
Right	Nationalist-secular right	Likud, Yisrael Beiteinu, Religious Zionism
Haredi	Ultra-Orthodox religious	Shas, United Torah Judaism
Center	Centrist parties	Yesh Atid, Blue and White, Kahol Lavan
Left	Labor-left	Labor, Meretz
Arab	Arab-majority parties	Joint List, Ra'am, Balad

This mapping enables the BlocShares feature representation, which aggregates raw party vote shares into five interpretable bloc dimensions.

4.4 Data Processing Pipeline

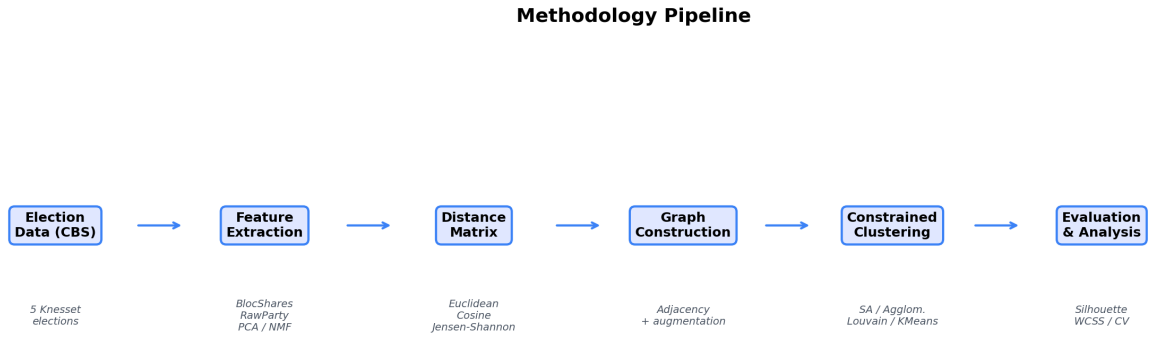

Figure 1: End-to-end methodology pipeline from raw election data to evaluated canton partitions.

The data processing pipeline (implemented in `src/data/`) handles: 1. **Loading** raw election Excel files and geographic shapefiles. 2. **Name normalization** for cross-dataset matching (handling Hebrew spelling variations, hyphenation, special characters). 3. **Municipality dissolution** from locality-level to municipality-level geographic polygons. 4. **Feature computation** from raw vote counts to vote share vectors.

All processing is reproducible from raw data files through the project's Jupyter notebooks (`notebooks/01_data_exploration.ipynb` through `notebooks/04_political_features.ipynb`).

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5. Methodology

5.1 Feature Representations

We evaluate four feature representations for municipality political profiles:

BlocShares (11 features): For each municipality, compute the mean and standard deviation of each bloc's vote share across all five elections, plus the average voter count. This produces an 11-dimensional vector: [right_avg, right_std, haredi_avg, haredi_std, center_avg, center_std, left_avg, left_std, arab_avg, arab_std, avg_votes]. The inclusion of temporal standard deviations captures voting stability, and the voter count helps balance canton populations during clustering.

RawParty (high-dimensional): The full vector of party-level vote shares, averaged across elections. This preserves fine-grained political distinctions but is high-dimensional and noisy due to small parties.

PCA_5 (5 features): Five principal components extracted from the RawParty representation via Principal Component Analysis. This captures the dominant axes of political variation while reducing dimensionality.

NMF_5 (5 features): Five components from Non-negative Matrix Factorization of the RawParty representation. NMF produces additive, non-negative components that can be interpreted as political archetypes (e.g., an "Arab voting pattern" component).

5.2 Distance Metrics

Three distance metrics are used to measure political dissimilarity between municipalities:

Euclidean Distance: The standard L2 norm between feature vectors. Sensitive to magnitude differences, which means municipalities with larger voter counts may dominate distance computations when voter count is included as a feature.

Cosine Distance: Measures the angular distance between feature vectors (1 - cosine similarity). Invariant to vector magnitude, focusing on the *direction* (relative proportions) rather than the *scale* of political features. This is particularly appropriate for vote share vectors where the absolute number of voters is less important than voting patterns.

Jensen-Shannon Distance: The square root of the Jensen-Shannon divergence [18], a symmetric, bounded measure derived from the Kullback-Leibler divergence. Particularly well-suited for probability distributions such as vote share vectors. The underlying divergence is:

$$\mathrm{JSD}(P\|Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M), \quad \text{where } M = \frac{1}{2}(P+Q)$$

Our implementation uses $d(P,Q) = \sqrt{\mathrm{JSD}(P\|Q)}$, which is a proper metric (satisfies the triangle inequality) [17].

5.3 Graph Construction and Preprocessing

Raw Adjacency Graph: We construct a contiguity graph $G = (V, E)$ where municipalities are nodes and edges connect municipalities that share a geographic boundary (detected via polygon intersection using GeoPandas and Shapely [22]). The raw graph has 234 nodes and 516 edges but is *disconnected* - some municipalities are geographic isolates (enclaves, exclave communities, or municipalities with no shared boundaries due to polygon precision issues).

Graph Preprocessing: To ensure a connected graph (required for contiguous clustering), we apply three preprocessing steps:

1. **Virtual edges for isolates:** For each isolated node (degree 0), connect it to its $K = 3$ nearest neighbors in feature space. $K = 3$ balances ensuring connectivity with avoiding spurious long-distance edges. This ensures all municipalities participate in the clustering.

2. **Enclave edges:** For municipalities where a single bloc exceeds 70% of the municipality's own vote share (political enclaves), add edges connecting all enclaves of the same bloc type. The 70% threshold identifies municipalities where a single bloc overwhelmingly dominates. This prevents small enclaves from being forced into politically dissimilar cantons.

3. **Bridge edges for disconnected components:** For any remaining disconnected components after steps 1-2, connect each component to the largest component via an edge between the most politically similar pair of municipalities. This ensures full graph connectivity.

The augmented graph has 234 nodes and 2,208 edges and is fully connected. During clustering, the graph is subsetted to the 229 analysis municipalities.

5.4 Clustering Algorithms

Simulated Annealing (SA): Our primary algorithm uses SA to optimize a multi-objective cost function over the partition space. Starting from a seed-based initialization (selecting the most politically extreme municipality per bloc as seeds, then greedily growing cantons with balance constraints), the algorithm proposes moves (reassigning a border municipality to an adjacent canton) and accepts or rejects them based on the Metropolis criterion. The cost

function weights three objectives: - Homogeneity (weight 0.4): Size-weighted mean within-canton feature variance. - Population balance (weight 0.4): Coefficient of variation of canton populations. - Compactness (weight 0.2): Graph cut ratio – fraction of inter-canton edges over total edges.

Parameters: initial temperature $T_0 = 1.0$, cooling rate 0.9995, maximum 50,000 iterations, following standard SA practice [9]. These parameters were sufficient for convergence in preliminary tests. Contiguity is enforced by rejecting moves that disconnect any canton. Note that SA uses a deterministic seed-based *initialization* (providing a reproducible starting point) but stochastic *optimization* (random move proposals with probabilistic acceptance), so results may vary with different random seeds. Each configuration was executed once; the stability analysis in Section 7 partially addresses sensitivity by evaluating SA across five different election datasets.

Agglomerative Clustering: Average-linkage with contiguity constraints [6] - starting from 229 singleton clusters, iteratively merge the two adjacent clusters with the smallest average pairwise distance. The contiguity constraint ensures only adjacent clusters merge, maintaining geographic connectedness. This deterministic algorithm produces a dendrogram that can be cut at any level to obtain K cantons.

K-Means Baseline: Standard K-Means clustering [5] in feature space without geographic contiguity constraints. Serves as a baseline to measure the cost of enforcing contiguity - cantons may be geographically disconnected.

Louvain Community Detection: The Louvain algorithm [8] maximizes modularity [16] on the adjacency graph. Edge weights are set based on political similarity (inverse distance). Louvain determines its own number of communities rather than accepting a target K, serving as a topology-driven baseline.

5.5 Evaluation Metrics

All algorithms were implemented in Python using scikit-learn [20] for K-Means and silhouette computation, NetworkX [21] for graph operations and Louvain community detection, and SciPy [22] for distance computations including Jensen-Shannon divergence.

Within-Canton Sum of Squares (WCSS): Measures total political variance within cantons. Computed as the sum of squared distances from each municipality's feature vector to its canton centroid (unweighted). Lower WCSS indicates more politically homogeneous cantons.

Silhouette Score [13]: For each municipality, measures how well it fits in its assigned canton versus the nearest alternative. Ranges from -1 to +1; higher values indicate better-defined clusters.

Population Balance (CV): Coefficient of variation of canton voter populations. Lower CV indicates more balanced cantons. We also report the max-to-min population ratio.

Contiguity: Number of cantons that are disconnected in the adjacency graph. Should be 0 for all spatially-constrained algorithms.

Dominant Margin: For each canton, the percentage-point lead of the dominant political bloc over the second-highest bloc. Higher margins indicate more politically distinctive cantons.

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6. Experimental Results

6.1 Experimental Setup

We conducted a systematic grid search over 264 configurations:

Table 3: Experimental grid dimensions.

Dimension	Values	Count
Representation	BlocShares, RawParty, PCA_5, NMF_5	4
Distance Metric	Euclidean, Cosine, Jensen-Shannon	3
Algorithm	SA, Agglomerative, Louvain, KMeans (baseline)	4
K (cantons)	3, 5, 7, 10, 15, 20	6
Total		**264**

Note: Not all metric-representation combinations are applicable. Jensen-Shannon requires non-negative values and is compatible with BlocShares, RawParty, and NMF_5, but not PCA_5 (which may produce negative values); PCA_5 uses only Euclidean and Cosine metrics. All 264 configurations executed successfully with 0 failures.

6.2 Key Findings

Representation Comparison: BlocShares consistently outperforms other representations across most metrics and algorithms. Its domain-informed aggregation of party votes into five blocs captures the politically meaningful dimensions while reducing noise. PCA_5 and NMF_5 perform comparably to each other but below BlocShares. RawParty, despite containing the most information, suffers from the curse of dimensionality [28], consistent with the known limitations of high-dimensional clustering [5].

Distance Metric Comparison: Euclidean distance produces the highest silhouette score overall (0.905 with BlocShares/Agglomerative at K=3). KMeans baseline achieves 0.858 at K=3 identically across all three metrics (Euclidean, Cosine, Jensen-Shannon), indicating that BlocShares features are robust to metric choice for unconstrained clustering. Jensen-Shannon performs well with RawParty shares but offers no clear advantage over Euclidean or Cosine with BlocShares.

Algorithm Comparison: - **Agglomerative** achieves the highest silhouette scores (up to 0.905) but can produce severe population imbalance, sometimes placing 225+ municipalities in a single canton. - **Simulated Annealing** produces the most balanced cantons due to its explicit population balance objective, at the cost of slightly lower silhouette scores. SA is the

preferred algorithm when population balance matters. - **Louvain** ignores the target K parameter and finds its own natural community count. Its partitions are perfectly stable across elections but may not match the desired K. - **KMeans (baseline)** achieves competitive silhouette scores but produces geographically disconnected cantons, confirming the necessity of contiguity constraints.

Effect of K: WCSS decreases monotonically with increasing K (as expected - more cantons means smaller, more homogeneous clusters). Silhouette scores [13] peak at low K values (K=3 or K=5) for most configurations, suggesting that Israel's political geography is naturally structured into a small number of macro-regions.

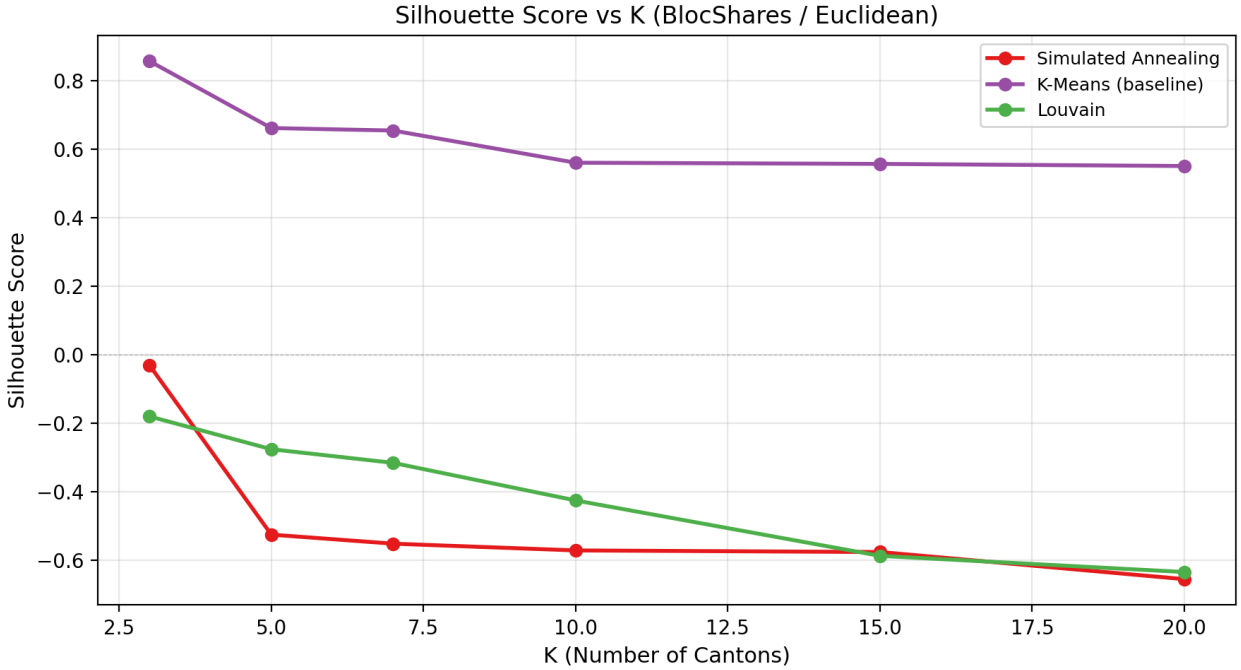

Figure 2: Silhouette score vs K for BlocShares/Euclidean across all four algorithms.

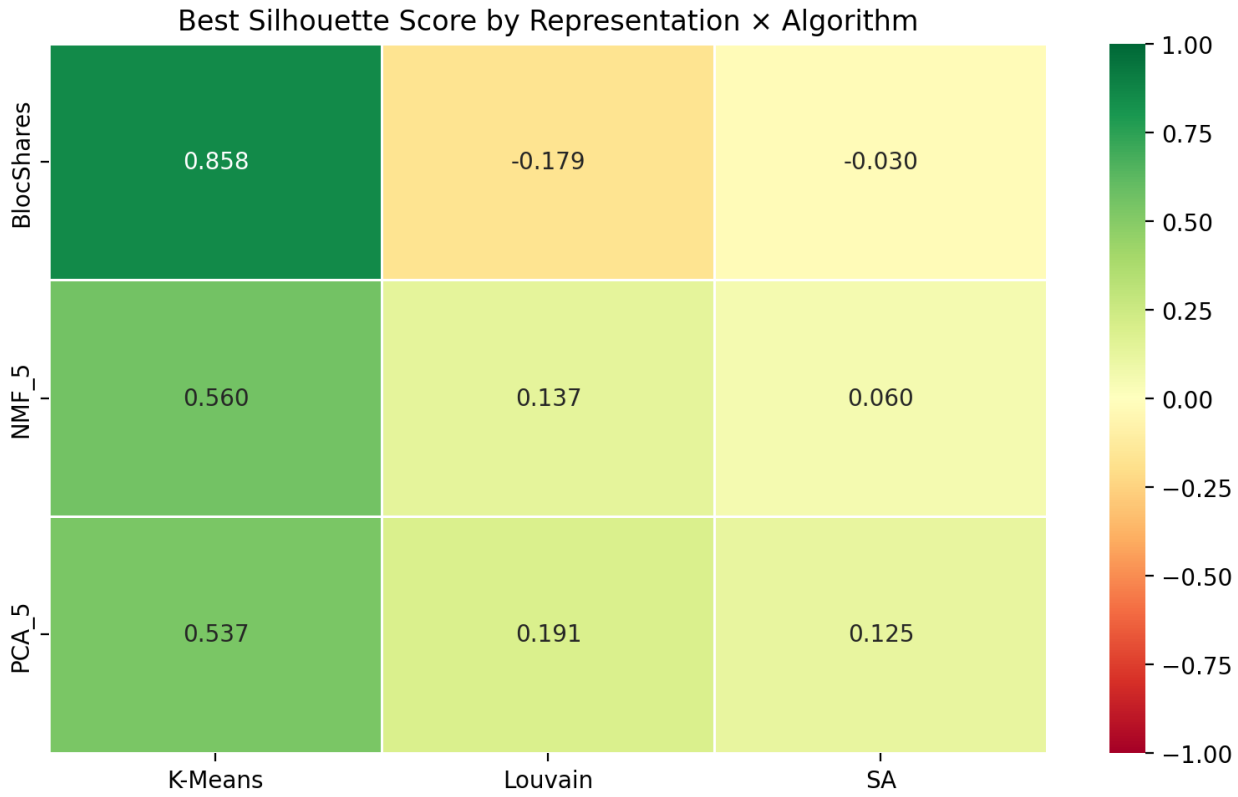

Figure 3: Best silhouette score achieved by each representation × algorithm combination.

6.3 Best Configurations

Table 4: Top five configurations ranked by silhouette score.

Rank	Repr	Metric	Algorithm	K	Silhouette	Pop CV
1	BlocShares	Euclidean	Agglomerative	3	0.905	1.13
2	BlocShares	Euclidean	KMeans	3	0.858	0.48
3	NMF_5	Euclidean	Agglomerative	3	0.542	1.14
4	BlocShares	Cosine	SA	5	-0.524	0.55
5	BlocShares	Euclidean	Louvain	auto	-0.179	0.64

The best overall configuration depends on which objectives are prioritized. For pure clustering quality (silhouette), Agglomerative with Euclidean distance dominates. KMeans achieves a high silhouette (0.858) but produces disconnected cantons. For balanced partitions that also maintain contiguity, SA with BlocShares/Cosine at K=5 is the preferred choice despite its negative silhouette, which reflects the extreme population imbalance between cantons rather than poor political separation.

6.4 Primary Result: K=5 Simulated Annealing Partition

We select K=5 as the primary result for interpretive analysis. While K=3 achieves higher silhouette (0.905 with Agglomerative), it produces only three coarse-grained regions with severe population imbalance (CV=1.13). K=5 with SA provides finer political resolution – distinguishing the Arab bloc, Druze enclave, coastal center, urban core, and southern periphery – while maintaining contiguity and reasonable balance.

Table 5: K=5 canton profiles (BlocShares / Cosine / SA). Bloc values are mean vote share (%).

Canton	Munis	Voters	Right%	Haredi%	Center%	Left%	Arab%
0 – RIGHT South	27	623,179	44.5	14.6	26.1	10.7	0.7
1 – RIGHT Center	18	1,120,796	35.6	25.2	29.4	7.5	0.3
2 – CENTER Coastal	62	1,169,734	39.3	6.3	39.5	11.2	1.3
3 – LEFT North	1	4,381	21.4	0.6	28.5	44.0	4.6
4 – ARAB North	121	1,005,988	19.9	4.0	17.0	7.4	49.7

Evaluation metrics for this partition: - Cantons: 5 - Population ratio (max/min): 267x - Population CV: 0.554 - Average within-canton standard deviation: 3,449 - Average dominant margin: 11.1% - Silhouette: -0.524 - Disconnected cantons: 0 (all contiguous)

The negative silhouette score (-0.524) requires careful interpretation. Silhouette is computed in the 11-dimensional BlocShares feature space using Cosine distance, and is heavily influenced by the extreme size imbalance: Canton 3 contains a single municipality (Beit Jann, 4,381 voters) while Canton 4 contains 121 municipalities. The single-municipality Canton 3 has no meaningful within-canton distance, which distorts the aggregate silhouette. Individual canton silhouette scores are not separately reported because the single-member Canton 3 renders per-canton analysis uninformative – the aggregate score is dominated by the size imbalance rather than political separation quality. By contrast, the K=3 Agglomerative partition achieves silhouette 0.905 but at the cost of placing 225+ municipalities in one canton (CV=1.13). The K=5 SA partition was selected as the primary result because it balances interpretability, contiguity, and population balance – objectives not captured by silhouette alone. Despite the negative silhouette, the cantons are politically coherent: each has a clearly dominant bloc, and the partition aligns with known political-geographic divisions (see Section 8).

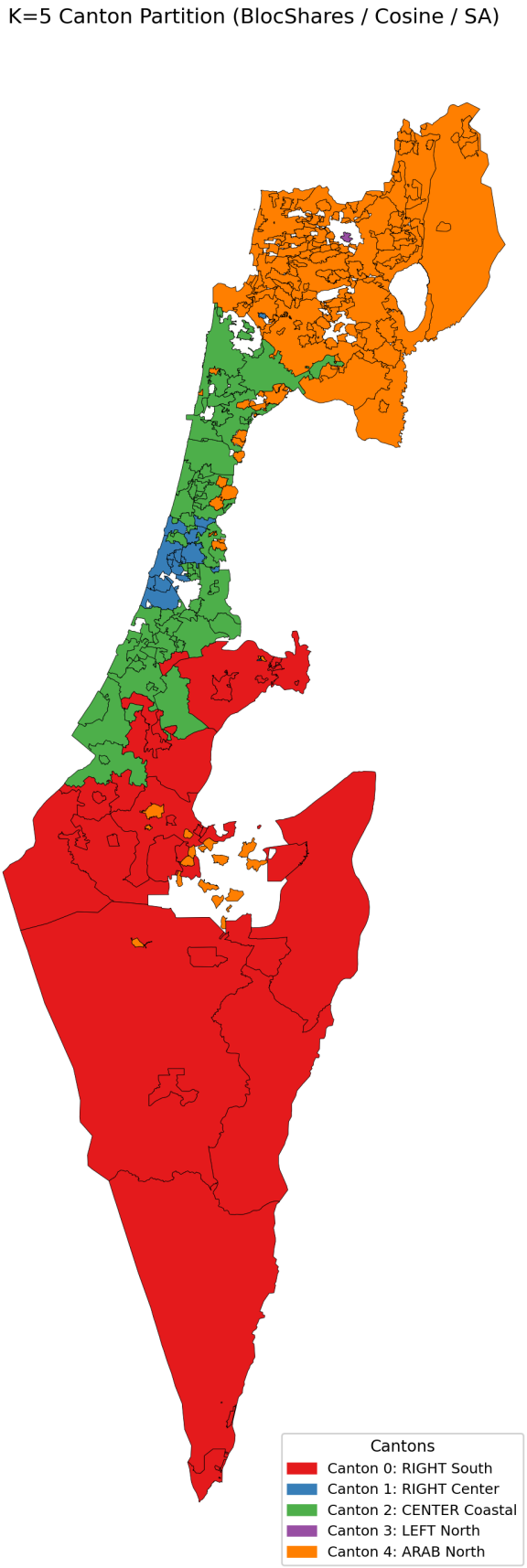

Figure 4: Geographic visualization of the K=5

SA canton partition.

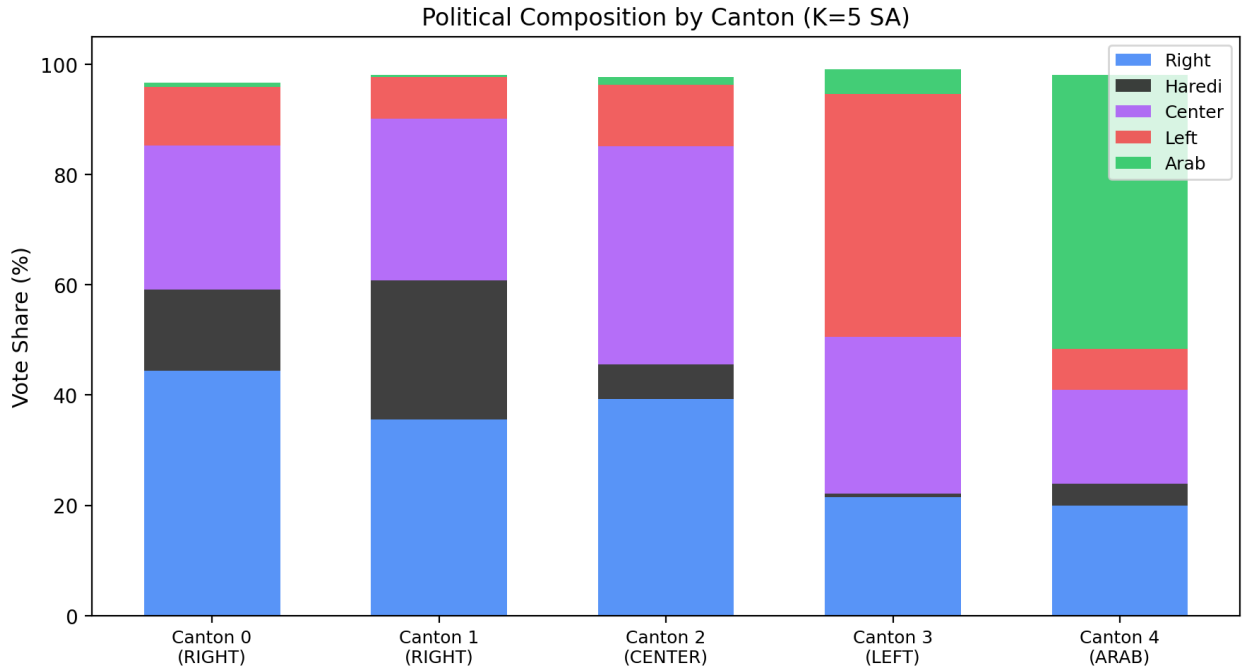

Figure 5: Political bloc composition of each canton in the K=5 SA partition.

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7. Stability Analysis

7.1 Methodology

To assess whether canton boundaries are structurally stable or sensitive to election-specific noise, we perform temporal stability analysis. For a fixed configuration, we run clustering independently on each of the five elections and measure pairwise agreement between the resulting partitions using:

- **Adjusted Rand Index (ARI) [14]:** Measures partition similarity corrected for chance. ARI = 1.0 indicates identical partitions; ARI = 0 indicates random agreement.
- **Normalized Mutual Information (NMI) [15]:** Information-theoretic measure of partition similarity. NMI = 1.0 indicates identical partitions.

We tested six representative configurations spanning different representation-metric-algorithm combinations.

7.2 Results

Table 6: Cross-election stability for six representative configurations.

Repr	Metric	Algorithm	Mean ARI	Std ARI	Mean NMI	Std NMI
BlocShares	Euclidean	Louvain	1.000	0.000	1.000	0.000
BlocShares	Cosine	Agglomerative	0.954	0.059	0.945	0.071
NMF_5	Euclidean	SA	0.616	0.233	0.682	0.155
BlocShares	Euclidean	SA	0.554	0.113	0.602	0.095
RawParty	Cosine	SA	0.451	0.120	0.550	0.099
PCA_5	Euclidean	SA	0.360	0.137	0.466	0.123

7.3 Interpretation

Deterministic algorithms produce highly stable partitions. Louvain achieves *perfect* stability (ARI = 1.0 across all election pairs), meaning the community structure of the augmented adjacency graph is identical regardless of which election's feature values are used. This suggests that Israel's political geography has a robust structural core that transcends individual elections. Agglomerative clustering also produces near-perfect stability (ARI = 0.954).

SA stability varies by representation. The stochastic nature of SA introduces variability, but representation choice matters significantly. NMF produces the most stable SA partitions (ARI = 0.616), followed by BlocShares (0.554) and RawParty (0.451). PCA produces the least stable partitions (0.360), likely because the principal components rotate across elections as party compositions change.

Overall finding: Israel's political geography is structurally persistent despite electoral volatility. The five elections, though producing different coalition outcomes, exhibit remarkably similar spatial patterns at the municipality level - consistent with the observation that political preferences are strongly spatially autocorrelated [30] and demographically determined.

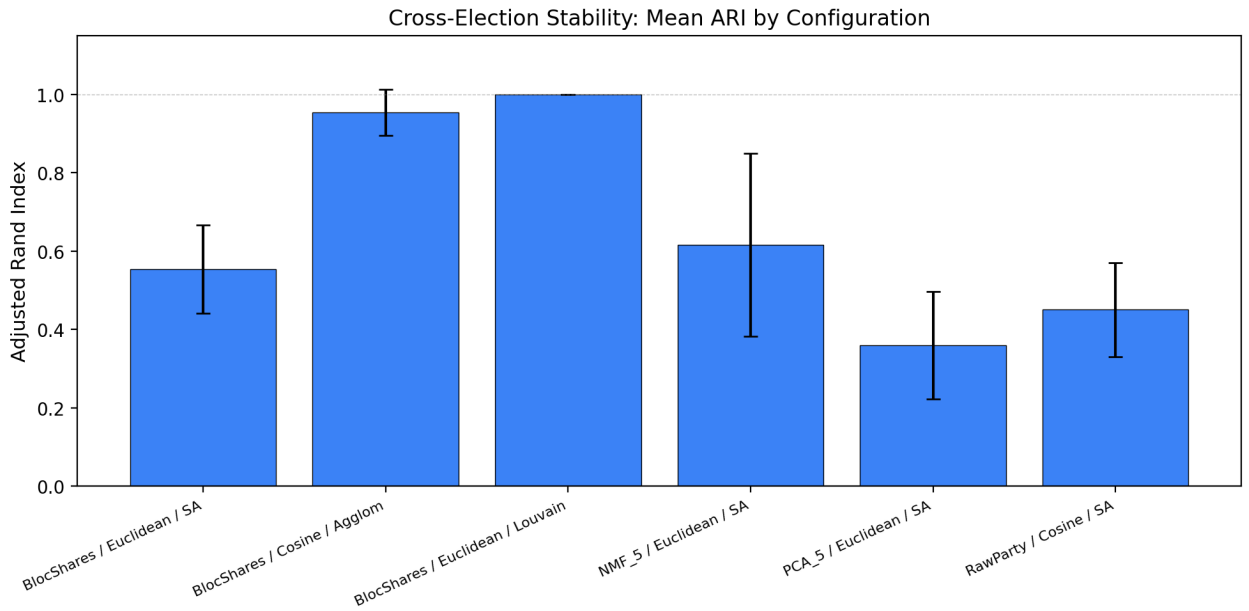

Figure 6: Mean Adjusted Rand Index across election pairs for each configuration, with error bars showing standard deviation.

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8. Case Studies

8.1 The K=5 Partition in Detail

Our primary K=5 SA partition divides Israel into five politically coherent regions. We examine five case study areas to assess whether the algorithmic cantons align with known political geography.

8.2 Greater Tel Aviv (Gush Dan)

The Greater Tel Aviv metropolitan area is split between **Canton 1** (RIGHT Center) and **Canton 2** (CENTER Coastal):

Table 7: Greater Tel Aviv canton assignments.

Municipality	Canton	Dominant Bloc
Tel Aviv-Yafo	1	Center
Ramat Gan	1	Center
Givatayim	1	Center
Bnei Brak	1	Haredi
Holon	1	Right
Bat Yam	1	Right
Petah Tikva	1	Right
Rishon LeZion	1	Right
Herzliya	1	Center

Municipality	Canton	Dominant Bloc
Ra'anana	2	Center
Kfar Saba	1	Center
Netanya	2	Right

Analysis: The core Gush Dan municipalities (Tel Aviv, Ramat Gan, Givatayim, Holon, Bat Yam, Petah Tikva) cluster together in Canton 1. This canton includes Bnei Brak - Israel's most prominent ultra-Orthodox city (Haredi-dominant) - alongside center-leaning Tel Aviv and right-leaning suburbs, reflecting the algorithm's treatment of the entire urban core as a single political unit. Canton 1 exhibits notable internal heterogeneity: it contains both Bnei Brak (Haredi-dominant) and Tel Aviv (Center-dominant), reflecting the geographic contiguity constraint which groups adjacent municipalities despite differing political profiles. Canton 2 (the coastal belt) captures the wider Sharon region municipalities (Ra'anana, Netanya) that are geographically distinct from the urban core – Ra'anana is center-leaning while Netanya leans right, but both share the coastal belt's moderate political profile.

8.3 Jerusalem Region

Table 8: Jerusalem region canton assignments.

Municipality	Canton	Dominant Bloc
Jerusalem	0	Haredi
Beit Shemesh	0	Haredi
Mevasseret Zion	0	Right
Modi'in-Maccabim-Re'ut	2	Center

Analysis: Jerusalem is placed in **Canton 0** (RIGHT South), grouped with the southern periphery towns (Beer Sheva, Dimona, Netivot, Ofakim) and Beit Shemesh. Both Jerusalem and Beit Shemesh are Haredi-dominant municipalities, reflecting their ultra-Orthodox-heavy voting patterns, which are politically more similar to the religious periphery than to the secular center. Modi'in, as a secular commuter town, is assigned to Canton 2 (CENTER Coastal) rather than the Jerusalem bloc.

8.4 Arab Towns

Table 9: Arab-majority municipality canton assignments.

Municipality	Canton	Dominant Bloc
Nazareth	4	Arab
Umm al-Fahm	4	Arab
Shfar'am	4	Arab
Tamra	4	Arab

Municipality	Canton	Dominant Bloc
Sakhnin	4	Arab
Tayibe	4	Arab
Baqa al-Gharbiyye	4	Arab
Rahat	4	Arab

Analysis: All Arab-majority municipalities are unified in **Canton 4** (ARAB North). This is the algorithm's clearest and most robust finding: the Arab voting pattern is so distinctive (49.7% Arab-party vote share, dramatically different from all other cantons) that Arab towns form a cohesive political bloc regardless of geographic location [25]. Bedouin towns in the Negev (Rahat, Hura, Laqiye, Tel Sheva) are grouped with northern Arab towns despite the geographic distance, connected through the augmented graph's virtual edges.

8.5 Haifa and the Krayot

Table 10: Haifa metropolitan area canton assignments.

Municipality	Canton	Dominant Bloc
Haifa	4	Right
Kiryat Bialik	4	Right
Kiryat Motzkin	4	Right
Kiryat Ata	4	Right
Kiryat Yam	4	Right
Nesher	4	Right

Analysis: Haifa and its satellite cities (the "Krayot") are assigned to **Canton 4** (ARAB North) despite being individually Right-dominant. This is the most heterogeneous canton assignment in the K=5 partition: these Jewish-majority cities are geographically embedded within the predominantly Arab northern region. The algorithm's contiguity constraint pulls them into the Arab canton because their geographic neighbors are overwhelmingly Arab towns. Haifa itself has a significant Arab population (~8%), further bridging the connection.

8.6 Southern Periphery

Table 11: Southern periphery canton assignments.

Municipality	Canton	Dominant Bloc
Beer Sheva	0	Right
Dimona	0	Right
Netivot	0	Haredi
Ofakim	0	Right
Arad	0	Right

Municipality	Canton	Dominant Bloc
Eilat	0	Right
Yeruham	0	Right
Mitzpe Ramon	0	Right

Analysis: The southern periphery forms a cohesive **Canton 0** (RIGHT South) with strong right-wing and Haredi voting patterns (44.5% Right, 14.6% Haredi). This aligns with the well-documented "periphery vote" phenomenon in Israeli politics [10, 25], where southern development towns consistently support right-wing parties. Netivot stands out as Haredi-dominant within the otherwise right-leaning canton.

8.7 Comparison to Administrative Districts

We compared our K=5 political cantons to Israel's official administrative districts (based on CBS locality data). The Adjusted Rand Index [14] between the canton partition and the administrative district partition is **ARI = 0.435**.

For reference, ARI = 0 indicates random agreement and ARI = 1 indicates identical partitions. This moderate overlap indicates that political cantons partially coincide with administrative districts but diverge significantly in key ways: - Administrative districts follow geographic boundaries (north, center, south), while political cantons follow ideological lines (Arab vs. Jewish, religious vs. secular, periphery vs. core). - The Arab canton (Canton 4) spans from the Galilee in the north to the Negev in the south, cutting across three administrative districts. - Jerusalem is administratively its own district but politically groups with the southern periphery.

This confirms that **political cantons reflect ideological geography rather than administrative geography**.

---

9. Limitations and Future Work

9.1 Limitations

Population Imbalance: The K=5 SA partition produces significant population imbalance. Canton 4 (ARAB North) contains 121 of 229 municipalities (53%), while Canton 3 (LEFT North) contains only 1 municipality (Beit Jann, a Druze village with 4,381 voters). Increasing the population balance weight in the SA cost function would improve balance but reduce political homogeneity.

Single-Municipality Canton: Canton 3 (Beit Jann) forms a single-municipality canton because its political profile - left-leaning Druze - is sufficiently distinct from all neighbors. While politically coherent, this is not a meaningful "canton" in any practical sense. Future work could enforce minimum canton sizes.

SA Single-Run Results: Simulated Annealing is a stochastic algorithm, and the results reported for each SA configuration are from a single run with a deterministic seed-based initialization. While the seed-based initialization reduces variance by providing a consistent starting point, different random seeds or initialization strategies could yield different partitions. Ideally, each SA configuration would be run multiple times to report mean and variance of evaluation metrics. Our stability analysis (Section 7) partially addresses this by evaluating SA on five different election datasets, revealing the sensitivity of SA results to input data variation.

Ecological Fallacy: Our analysis operates at the municipality level, aggregating individual votes. This means we cannot infer individual-level political preferences from municipality-level patterns - a well-known limitation in spatial analysis called the ecological fallacy [19]. For example, a right-leaning municipality may contain left-leaning neighborhoods, but our municipality-level data cannot capture such within-municipality variation. Our partition describes municipal-level political geography, not individual voter preferences.

Static Analysis: We analyze election results as fixed snapshots without modeling voter migration, demographic shifts, or political realignment over time. Our stability analysis shows structural persistence across five elections, but longer-term changes (such as growth of specific religious or ethnic communities) are not captured.

Data Coverage: We include only the 229 municipalities that are consistently matchable across all five elections and geographic databases. Small localities, kibbutzim, and moshavim that are part of regional councils (administrative units encompassing multiple small localities) are aggregated at the regional council level, potentially masking within-council political heterogeneity.

Contiguity Approximation: The augmented graph includes virtual edges for isolate municipalities and enclaves, which means some "contiguous" cantons are not strictly geographically connected but rather connected through feature-space proximity. This is a pragmatic compromise to handle Israel's complex municipal geography.

KMeans Baseline Contiguity: The KMeans baseline operates purely in feature space without geographic constraints, meaning its cantons may be geographically disconnected. It is included as a reference point to quantify the trade-off between clustering quality and contiguity enforcement.

9.2 Future Work

Enhanced Web Application: The current Streamlit-based explorer allows interactive exploration of all 264 configurations. Future work could extend this with React + Leaflet for richer interactivity, side-by-side canton comparisons, and animated transitions between configurations.

Finer Geographic Resolution: Extend the analysis to statistical area or neighborhood level within large cities, capturing within-city political variation (e.g., north vs. south Tel Aviv).

Coalition Simulation: Use canton-level political profiles to simulate coalition formation under hypothetical regional representation systems.

Cross-Country Comparison: Apply the same constrained spatial clustering methodology to other countries with spatial political polarization (e.g., the United States red/blue state divide, Belgian linguistic-political divisions).

Temporal Dynamics: Model how canton boundaries shift over time as demographics change, incorporating census data and immigration statistics.

Minimum Canton Size Constraint: Add a hard constraint on minimum canton population to prevent single-municipality cantons.

Parameter Sensitivity Analysis: A systematic sensitivity analysis of SA cost function weights, graph preprocessing thresholds (enclave threshold, nearest-neighbor K), and SA temperature schedule parameters would further strengthen the methodology and quantify robustness of results to parameter choices.

---

10. Conclusion

This project successfully demonstrates that constrained spatial clustering can partition Israeli municipalities into politically coherent, geographically contiguous cantons that align with known political-demographic divisions. Our systematic experimental framework - evaluating 264 configurations across four representations, three distance metrics, four algorithms, and six values of K - provides comprehensive evidence that:

1. **Israel's political geography is structurally robust.** The BlocShares representation, which captures five major political dimensions (Right, Haredi, Center, Left, Arab), consistently produces the highest-quality clusters. Temporal stability analysis shows near-perfect partition agreement across five elections for deterministic algorithms (ARI up to 1.0).

2. **The Arab-Jewish divide is the dominant political-geographic cleavage.** In virtually all configurations, Arab-majority municipalities form a distinct, cohesive cluster, reflecting the unique Arab voting pattern that is quantitatively dissimilar from all Jewish-majority voting patterns.

3. **Political cantons differ from administrative districts.** With ARI = 0.435 [14] between our K=5 cantons and Israel's administrative districts, we confirm that political geography follows ideological lines (religious vs. secular, periphery vs. core, Arab vs. Jewish) rather than administrative boundaries.

4. **Multi-objective optimization is essential.** The trade-off between political homogeneity, population balance, and geographic compactness requires explicit multi-objective methods. Simulated Annealing, despite lower silhouette scores, produces the most practically useful partitions by balancing all three objectives.

5. **Distance metric choice matters.** Euclidean distance with BlocShares/Agglomerative at K=3 achieves the highest silhouette score (0.905), while Cosine and Jensen-Shannon perform comparably for some representations. The choice of metric interacts with the representation: magnitude-sensitive Euclidean works best with the already-normalized BlocShares representation.

This project bridges computer science, data science, geographic information systems, and political analysis, providing both a methodological contribution to constrained spatial clustering and an analytical tool for exploring Israel's political geography. The open-source codebase, modular architecture, and comprehensive experiment logs enable full reproducibility and extension.

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Appendix

A. Canton Member Lists (K=5 SA Partition)

Canton 0 - RIGHT South (27 municipalities): Sdot Negev, Dimona, Netivot, Merhavim, Ofakim, Beer Sheva, Omer, Eshkol, Bnei Shimon, Lakhish, Kiryat Gat, Mateh Yehuda, Mevasseret Zion, Beit Shemesh, Meitar, Lehavim, Shafir, Ramat Negev, Yeruham, Mitzpe Ramon, Tamar, Arad, HaArava HaTikhona, Hevel Eilot, Eilat, Jerusalem, Nahal Sorek

Canton 1 - RIGHT Center (18 municipalities): Bnei Brak, Rekhesim, Kiryat Ye'arim, El'ad, Ramat Gan, Petah Tikva, Or Yehuda, Givat Shmuel, Hod HaSharon, Herzliya, Kfar Saba, Givatayim, Tel Aviv-Yafo, Bat Yam, Holon, Azor, Rishon LeZion, Beit Dagan

Canton 2 - CENTER Coastal (62 municipalities): Daliyat al-Karmel, Isfiya, Hof HaSharon, Netanya, Ra'anana, Tel Mond, Kfar Shemaryahu, Even Yehuda, Kadima-Zoran, Lev HaSharon, Pardesiya, Kfar Yona, Drom HaSharon, Savyon, Kokhav Ya'ir, Ganei Tikva, Hevel Modi'in, Shoham, Gezer, Mazkeret Batya, Brenner, Gan Raveh, Gedarot, Be'er Tuvia, Hof Ashkelon, Gedera, Yehud-Monosson, Be'er Ya'akov, Gan Yavne, Bnei Ayish, Kiryat Ekron, Kiryat Ono, Sha'ar HaNegev, Nes Tziona, Yoav, Ramat HaSharon, Emek Hefer, Elyakhin, Menashe, Alona, Binyamina-Giv'at Ada, Zikhron Ya'akov, Hof HaCarmel, Pardes Hanna-Karkur, Yavne, Tirat Carmel, Rosh HaAyin, Or Akiva, Sderot, Harish, Modi'in-Maccabim-Re'ut, Megiddo, Yokne'am Illit, Hevel Yavne, Ramla, Hadera, Lod, Kiryat Malakhi, Rehovot, Ashkelon, Ashdod, Afula

Canton 3 - LEFT North (1 municipality): Beit Jann

Canton 4 - ARAB North (121 municipalities): Arraba, Nazareth, Umm al-Fahm, Sakhnin, Dabburiyya, Kaukab Abu al-Hija, Mashhad, Mazra'a, Kuseife, Tel Sheva, Segev Shalom, Ar'ara BaNegev, Sha'ab, Laqiye, Neve Midbar, Tur'an, Dir Hanna, Iksal, Jadeidi-Makr, Majd al-Krum, Kafr Yasif, Yafi', Ein Mahil, Bi'ina, Reina, Kabul, Hura, Bu'eine Nujeidat, Nahf, Basma, Jatt, I'billin, Ma'ale Iron, Kafr Kanna, Qalansuwa, Kafr Qara, Baqa al-Gharbiyye, Ar'ara, Jaljulia, Dir al-Asad, Zemer, Tamra, Tayibe, Shfar'am, Rahat, Kafr Manda, Al-Batuf, Ma'ilya, Ilut, Tira, Ma'alot-Tarshiha, Abu Snan, Kafr Bara, Al-Qasam, Jisr az-Zarqa, Magar, Ilabun, Bustan al-Marj, HaGilboa, Zvulun, Misgav, Rama, Kiryat Bialik, Kiryat Motzkin, Nahariya, Kiryat Ata, Karmi'el, Nesher, HaGalil HaTahton, Emek HaMa'ayanot, Ma'ale Yosef, Fassuta, Merom HaGalil, Gush Halav (Jish), Mevo'ot HaHermon, Yesod HaMa'ala, Kfar Tavor, Golan, Mateh Asher, Akko, Metula, Rosh Pinna,

Katzrin, Migdal, Shlomi, Migdal HaEmek, Kiryat Shmona, Kiryat Yam, Kfar Vradim, Abu Ghosh, Yanouh-Jat, Hatzor HaGlilit, Hurfeish, Peqi'in (Buqei'a), Beit She'an, Tiberias, Emek HaYarden, Kiryat Tiv'on, Emek Yizre'el, Ramat Yishai, Safed, Yavne'el, Yirka, Basmat Tab'un, Majdal Shams, Kafr Kama, HaGalil HaElyon, Haifa, Bir al-Maksur, Sa'ur, Ein Kinya, Mas'ada, Kisra-Sumei, Kafr Qasem, Ka'abiyye-Tabbash-Hajajre, Fureidis, Zarziir, Tuba-Zangariyye, Buq'ata, Julis, Ghajar

B. Software Architecture

```
src/
  config.py              - Project paths, constants, bloc column definitions
  data/
    loader.py            - Load election data, GeoJSON, municipality mappings
    processing.py        - Process raw elections, normalize names, match data
    representations.py   - BlocShares, RawParty, PCA, NMF feature builders
    distance_metrics.py  - Euclidean, Cosine, Jensen-Shannon distance classes
  graph/
    adjacency.py         - Build adjacency graph from geographic polygons
    preprocessing.py     - Virtual edges, enclave edges, component bridging
  clustering/
    base.py              - CantonAssignment data class, base clusterer interface
    simulated_annealing.py - SA clusterer with multi-objective cost function
    agglomerative.py     - Average-linkage with contiguity constraints
    spectral.py          - Louvain community detection (used); spectral clustering (impl
    baseline.py          - K-Means baseline (no contiguity constraints)
  evaluation/
    metrics.py           - WCSS, silhouette, population balance, contiguity
    stability.py         - Temporal stability (ARI/NMI across elections)
    experiment.py        - Experiment grid runner, result aggregation
  visualization/
    maps.py              - Static canton maps, interactive Folium maps
    charts.py            - Political composition, population balance, elbow plots

tests/                   - 73 unit tests covering all modules
notebooks/               - 6 Jupyter notebooks (exploration through experiments)
app/
  app.py                 - Streamlit main page (project overview, featured K=5 map)
  utils.py               - Shared data loading and visualization helpers
  pages/
    1_Canton_Map.py      - Interactive canton map explorer (all 264 configurations)
    2_Experiments.py     - Experiment results dashboard (heatmaps, line plots)
    3_Stability.py       - Cross-election stability analysis (ARI/NMI charts)
    4_About.py           - Project background and methodology overview
```

C. Reproducibility

Key Dependencies: Python 3.12, pandas, geopandas, scikit-learn, networkx, scipy, matplotlib, seaborn, plotly, streamlit, folium.

All results can be reproduced by running the Jupyter notebooks in sequence: 1. `01_data_exploration.ipynb` - Initial data inspection 2. `02_data_processing.ipynb` - Data cleaning and integration 3. `03_adjacency_graph.ipynb` - Graph construction and preprocessing 4. `04_political_features.ipynb` - Feature representation computation 5. `05_clustering.ipynb` - Primary SA clustering (K=5) 6. `06_experiments.ipynb` - Full 264-config grid search, stability analysis, case studies

Requirements: Python 3.11+, dependencies listed in `requirements.txt`. All source data is included in the `data/raw/` directory.